

TIMOTHY ODONGA

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EDUCATION

Doctor of Philosophy in Computer Science and Informatics Emory University	Aug. 2023 – May 2028 (expected)
Master of Science in Electrical and Computer Engineering Carnegie Mellon University	Aug. 2018 – Dec. 2019
Bachelor of Science in Electrical Engineering University of Southern California	Aug. 2014 – May 2016
Bachelor of Science in Physics Minor in Mathematics Gordon College	Aug. 2011 – May 2016

SELECTED EXPERIENCE

Graduate Research Assistant Emory University	August 2023 – to date Georgia, USA
- Conducting research to advance trustworthy machine learning (i.e., algorithmic fairness, interpretability) and AI alignment for automated clinical movement disorders assessment systems in the ViTAL lab and the Emory Brain Health 3D Motion Capture Laboratory. - Developing deep learning models for tremor detection, Freezing of Gait evaluation in Parkinson's Disease, and motor assessment in infant Cerebral Palsy using wearable and motion capture data.	
Data Scientist Dalberg Data Insights	January 2021 – June 2023 Kampala, Uganda
- Building algorithms and designing experiments to merge, investigate, and extract data to supply tailored reports to clients. - Using machine learning and statistical techniques to produce solutions to problems.	
Research Intern IBM Research	June 2019 – September 2019 Nairobi, Kenya
- Conducted research on algorithmic fairness of benchmark machine learning models used in dermatology. - Implemented experiments on deep learning models for image classification and segmentation for dermatology. - Contributed to a novel approach for evaluating skin tone bias in dermatology image datasets, found and reported that performance for benchmark skin disease diagnosis models is uniform across skin tone categories.	

PUBLICATIONS & PREPRINTS

- Timothy Odonga**, Jeanne M. Powell, Mark Saad, Richa Tripathi, Christine D. Esper, Stewart A. Factor, Hyeokhyen Kwon, and J. Lucas Mckay. "An Explainable Hierarchical Self Attention-Based Approach for Tremor Detection in the Time Domain." arXiv preprint arXiv:2606.00461 (2026).
- Timothy Odonga**, Jeanne M. Powell, Claire Bae, Yumeng Tan, Christine D. Esper, Stewart A. Factor, J. Lucas Mckay, and Hyeokhyen Kwon. "Evidence for Phenotype-Driven Disparities in Freezing of Gait Detection and Approaches to Bias Mitigation." arXiv preprint arXiv:2502.09626 (2025). (Under Review at European Journal for Neuroscience)
- Kinyanjui Newton M., **Timothy Odonga**, Celia Cintas, Noel CF Codella, Rameswar Panda, Prasanna Sattigeri, and Kush R. Varshney. "Fairness of Classifiers Across Skin Tones in Dermatology." In International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 320-329. Springer, Cham, 2020.
- Kinyanjui Newton M., **Timothy Odonga**, Celia Cintas, Noel C.F. Codella, Rameswar Panda, Prasanna Sattigeri, and Kush R. Varshney. "Estimating Skin Tone and Effects on Classification Performance in Dermatology Datasets." arXiv preprint arXiv:1910.13268 (2019).

SELECTED HONORS AND AWARDS

Lakshmi Subramanian CMU-Africa Student Excellence Award Awarded to a student that demonstrated leadership, academic excellence, & exceptional contributions to student life.	May 2020
IBM Great Minds Scholar & AI for Social Good Fellow Selected among students from Europe, the Middle East, & Africa to work alongside world-class scientists at IBM.	May 2019
Gordon College Graduation Honors Summa Cum Laude	May 2016
USC Dean's Transfer Merit Scholarship	May 2014
Jerrold L. McNatt Scholarship at Gordon College	April 2013
Sigma Pi Sigma National Honor Society for Physics	April 2013

CONFERENCES, INVITED TALKS, AND SUMMER SCHOOLS

- Poster to be presented at the American Medical Informatics Association Annual Symposium in November 2026 (Dallas, TX, USA) (Accepted).
- Poster presentation at the American Medical Informatics Association Annual Symposium in November 2025 (Atlanta, Georgia, USA).
- Poster presentation at the International Consortium for Freezing of Gait in June 2025 (Atlanta, Georgia, USA).
- Poster presentation at the Wearable Innovations for Health Equity Workshop in September 2024 (Atlanta, Georgia, USA).
- Poster presentation at the Gait and Clinical Movement Analysis Society (GCMAS) annual conference in June 2024 (Atlanta, Georgia, USA).
- Served as a diverse voices panel reviewer for the book “Trustworthy Machine Learning” written by Dr. Kush Varshney in February 2022.
- Invited talk titled, “Beyond Patterns 19: Fairness of Classifiers Across Skin Tones in Dermatology” at Friedrich Alexander Universität Erlangen-Nürnberg (FAU) in March 2021 (virtual).
- Poster and spotlight presentation at the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) in October 2020 (virtual).
- Completed the Interactive track of the Neuromatch Academy in July 2020 (virtual).
- Completed the IBRO-Simons Computational Neuroscience Imbizo summer school in Cape Town, South Africa, in January 2020 (Cape Town, South Africa).
- Poster and spotlight presentation at the Fair ML for Health workshop at the NeurIPS conference in December 2019 (Vancouver Canada).

SKILLS

Programming Languages: Python, R, JavaScript, C/C++, Matlab, Java

Libraries & Frameworks: PyTorch, LangChain, Pandas, OpenCV, Tidyverse, Scikit-learn, PySpark, NLTK, Tensorflow, JAX

Web Frameworks: Django, Fast API, Dash, React

Tools: R Studio, Git, Docker, AWS, GCP

TEACHING AND MENTORSHIP

- Graduate Teaching Assistant at Emory University: QTM-100 Introduction To Statistical Inference, Fall 2025; BMI-500 Introduction to Ethical Data Science and Informatics, Fall 2024; CS-224 Foundations of Computer Science, Spring 2024; CS-170 Introduction to Programming, Fall 2023
- Graduate Teaching Assistant at Carnegie Mellon University - Africa: ICT-04801 Cognitive Robotics, Spring 2020; ECE-18799 Applied Computer Vision, Fall 2024
- Mentees: Wendy Chau, BSc at Emory University; Sivesh Kannan, MSc at Georgia Institute of Technology; Yaqi Liu, MSc at Georgia Institute of Technology

LEADERSHIP AND SERVICE

- Treasurer of the Emory International InterVarsity Fellowship - Fall 2025.
- Voting Member on the Department of Biomedical Informatics (BMI) Awards Committee - Spring 2025.
- Mentor for the Black in AI Emerging Leaders in AI (ELAI) program - Fall 2025.
- Mentor with the Women in Machine Learning & Data Science (WiMLDS) Kampala chapter.
- Secretary of the Data Science Club at CMU-Africa 2018-2019.
- Executive member on the Gordon College International Students' Organization 2012- 2013.

REFERENCES

Can be provided upon request.